

# Tables

Table 1. Summary of the prompt conditions.

| Response Options | Image Information                                                                   | Prompts |
|------------------|-------------------------------------------------------------------------------------|---------|
| 0 or 1           | The input is described as a chest X-ray. Image absence is not explicitly disclosed. | A-F     |
| 0 or 1           | The prompt explicitly states that the chest X-ray image is unavailable.             | A-F     |
| 0, 1, or 2       | The prompt explicitly states that the chest X-ray image is unavailable.             | A-F     |
| 0, 1, or 2       | The input is described as a chest X-ray. Image absence is not explicitly disclosed. | A-F     |

Table 2. Comparison of MedGemma and Ministral Performance Across Prompts.

| Prompt | MedGemma F1 $\pm$ SD<br>[95% CI]    | Ministral F1 $\pm$ SD<br>[95% CI]   |
|--------|-------------------------------------|-------------------------------------|
| A      | 0.659 $\pm$ 0.063<br>[0.619, 0.702] | 0.593 $\pm$ 0.074<br>[0.509, 0.653] |
| B      | 0.659 $\pm$ 0.063<br>[0.619, 0.702] | 0.659 $\pm$ 0.041<br>[0.613, 0.692] |
| C      | 0.659 $\pm$ 0.063<br>[0.619, 0.700] | 0.633 $\pm$ 0.057<br>[0.577, 0.690] |
| D      | 0.629 $\pm$ 0.115<br>[0.541, 0.700] | 0.672 $\pm$ 0.112<br>[0.548, 0.756] |
| E      | 0.621 $\pm$ 0.118<br>[0.541, 0.695] | 0.642 $\pm$ 0.050<br>[0.594, 0.691] |
| F      | 0.684 $\pm$ 0.093<br>[0.623, 0.744] | 0.679 $\pm$ 0.016<br>[0.661, 0.695] |

Table 3. Effect of Response Format on Normalized Response-Distribution Entropy.

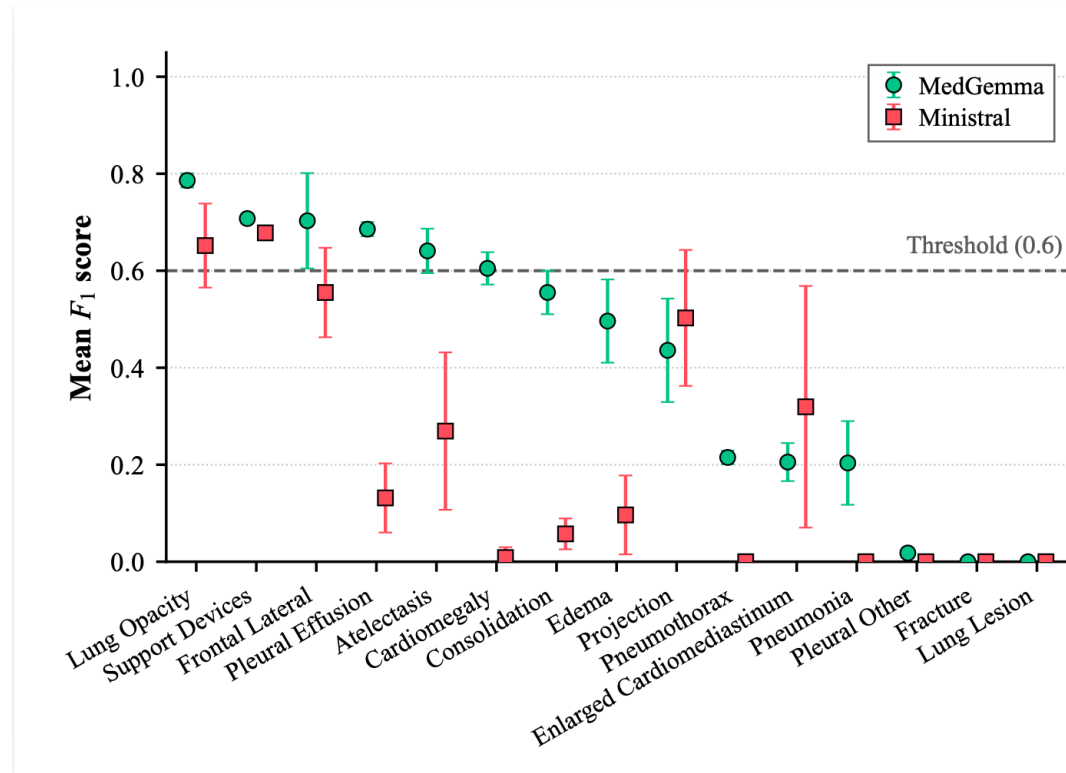
| Condition          | Model     | Binary | Three-Choice | Difference |
|--------------------|-----------|--------|--------------|------------|
| Normal             | MedGemma  | 0.126  | 0.102        | -0.024     |
| Normal             | Ministral | 0.125  | 0.316        | +0.191     |
| No Image (Unknown) | MedGemma  | 0.392  | 0.352        | -0.040     |
| No Image (Unknown) | Ministral | 0.083  | 0.173        | +0.090     |
| No Image (Known)   | MedGemma  | 0.704  | 0.469        | -0.235     |
| No Image (Known)   | Ministral | 0.031  | 0.139        | +0.108     |

**Table 4. Category-Level Classification Performance of MedGemma and Ministral.**

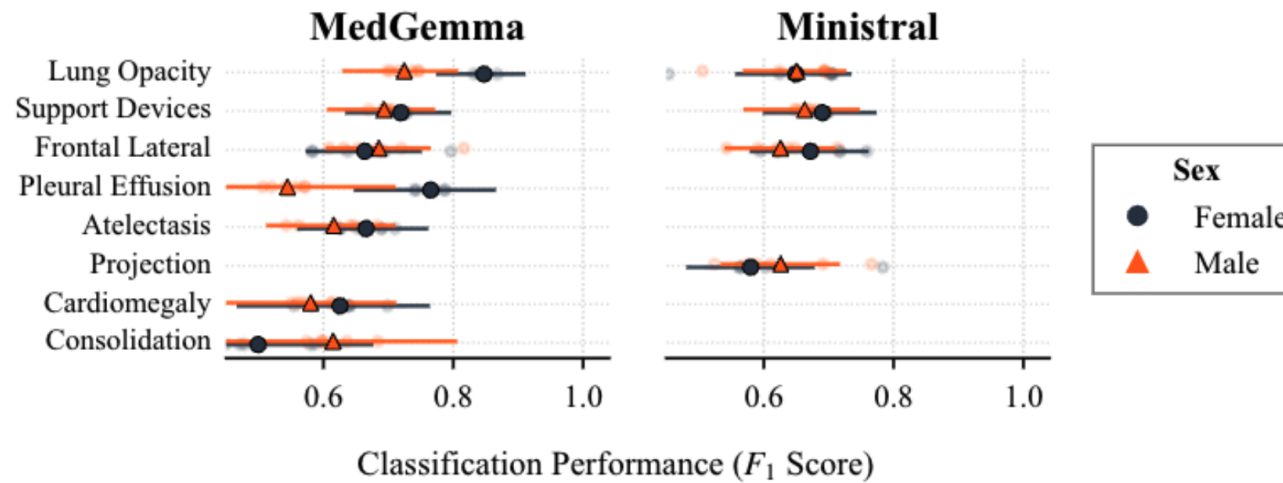
| Category                     | Mean F1 $\pm$ SD<br>[95% CI]        | Mean Acc $\pm$ SD<br>[95% CI]       |
|------------------------------|-------------------------------------|-------------------------------------|
| <b>MedGemma Performance</b>  |                                     |                                     |
| Frontal Lateral              | 0.712 $\pm$ 0.098<br>[0.686, 0.738] | 0.686 $\pm$ 0.074<br>[0.652, 0.700] |
| Projection                   | 0.433 $\pm$ 0.107<br>[0.396, 0.468] | 0.654 $\pm$ 0.134<br>[0.636, 0.679] |
| Enlarged Cardiomeastinum     | 0.207 $\pm$ 0.039<br>[0.169, 0.246] | 0.562 $\pm$ 0.008<br>[0.525, 0.576] |
| Cardiomegaly                 | 0.605 $\pm$ 0.033<br>[0.562, 0.664] | 0.788 $\pm$ 0.024<br>[0.769, 0.804] |
| Lung Opacity                 | 0.786 $\pm$ 0.015<br>[0.762, 0.809] | 0.650 $\pm$ 0.009<br>[0.625, 0.675] |
| Lung Lesion                  | 0.000 $\pm$ 0.000<br>[0.000, 0.000] | 0.900 $\pm$ 0.065<br>[0.884, 0.916] |
| Edema                        | 0.488 $\pm$ 0.086<br>[0.432, 0.543] | 0.777 $\pm$ 0.061<br>[0.756, 0.798] |
| Consolidation                | 0.575 $\pm$ 0.051<br>[0.538, 0.609] | 0.739 $\pm$ 0.020<br>[0.716, 0.763] |
| Pneumonia                    | 0.166 $\pm$ 0.068<br>[0.116, 0.217] | 0.778 $\pm$ 0.140<br>[0.871, 0.904] |
| Atelectasis                  | 0.638 $\pm$ 0.046<br>[0.608, 0.667] | 0.578 $\pm$ 0.077<br>[0.552, 0.603] |
| Pneumothorax                 | 0.214 $\pm$ 0.014<br>[0.112, 0.314] | 0.966 $\pm$ 0.005<br>[0.957, 0.975] |
| Pleural Effusion             | 0.687 $\pm$ 0.015<br>[0.647, 0.724] | 0.660 $\pm$ 0.007<br>[0.636, 0.685] |
| Pleural Other                | 0.016 $\pm$ 0.010<br>[0.003, 0.032] | 0.994 $\pm$ 0.101<br>[0.989, 0.997] |
| Fracture                     | 0.000 $\pm$ 0.000<br>[0.000, 0.000] | 0.956 $\pm$ 0.108<br>[0.945, 0.967] |
| Support Devices              | 0.707 $\pm$ 0.010<br>[0.683, 0.731] | 0.622 $\pm$ 0.018<br>[0.580, 0.630] |
| <b>Ministral Performance</b> |                                     |                                     |
| Frontal Lateral              | 0.561 $\pm$ 0.092<br>[0.525, 0.596] | 0.676 $\pm$ 0.078<br>[0.652, 0.700] |
| Projection                   | 0.515 $\pm$ 0.140<br>[0.475, 0.554] | 0.639 $\pm$ 0.095<br>[0.612, 0.667] |
| Enlarged Cardiomeastinum     | 0.388 $\pm$ 0.249<br>[0.350, 0.425] | 0.550 $\pm$ 0.030<br>[0.525, 0.576] |
| Cardiomegaly                 | 0.010 $\pm$ 0.021<br>[0.000, 0.024] | 0.706 $\pm$ 0.009<br>[0.682, 0.729] |
| Lung Opacity                 | 0.662 $\pm$ 0.087<br>[0.637, 0.686] | 0.523 $\pm$ 0.033<br>[0.496, 0.549] |
| Lung Lesion                  | 0.000 $\pm$ 0.000<br>[0.000, 0.000] | 0.976 $\pm$ 0.022<br>[0.968, 0.984] |
| Edema                        | 0.108 $\pm$ 0.080<br>[0.065, 0.155] | 0.777 $\pm$ 0.025<br>[0.756, 0.798] |

| Category         | Mean F1 $\pm$ SD<br>[95% CI]        | Mean Acc $\pm$ SD<br>[95% CI]       |
|------------------|-------------------------------------|-------------------------------------|
| Consolidation    | 0.062 $\pm$ 0.032<br>[0.031, 0.096] | 0.739 $\pm$ 0.080<br>[0.716, 0.763] |
| Pneumonia        | 0.000 $\pm$ 0.000<br>[0.000, 0.000] | 0.888 $\pm$ 0.053<br>[0.871, 0.904] |
| Atelectasis      | 0.310 $\pm$ 0.162<br>[0.269, 0.350] | 0.578 $\pm$ 0.059<br>[0.552, 0.603] |
| Pneumothorax     | 0.000 $\pm$ 0.000<br>[0.000, 0.000] | 0.966 $\pm$ 0.002<br>[0.957, 0.975] |
| Pleural Effusion | 0.144 $\pm$ 0.071<br>[0.104, 0.350] | 0.660 $\pm$ 0.050<br>[0.636, 0.685] |
| Pleural Other    | 0.000 $\pm$ 0.000<br>[0.000, 0.000] | 0.994 $\pm$ 0.005<br>[0.989, 0.997] |
| Fracture         | 0.000 $\pm$ 0.000<br>[0.000, 0.000] | 0.956 $\pm$ 0.026<br>[0.945, 0.967] |
| Support Devices  | 0.678 $\pm$ 0.009<br>[0.651, 0.703] | 0.605 $\pm$ 0.016<br>[0.580, 0.630] |

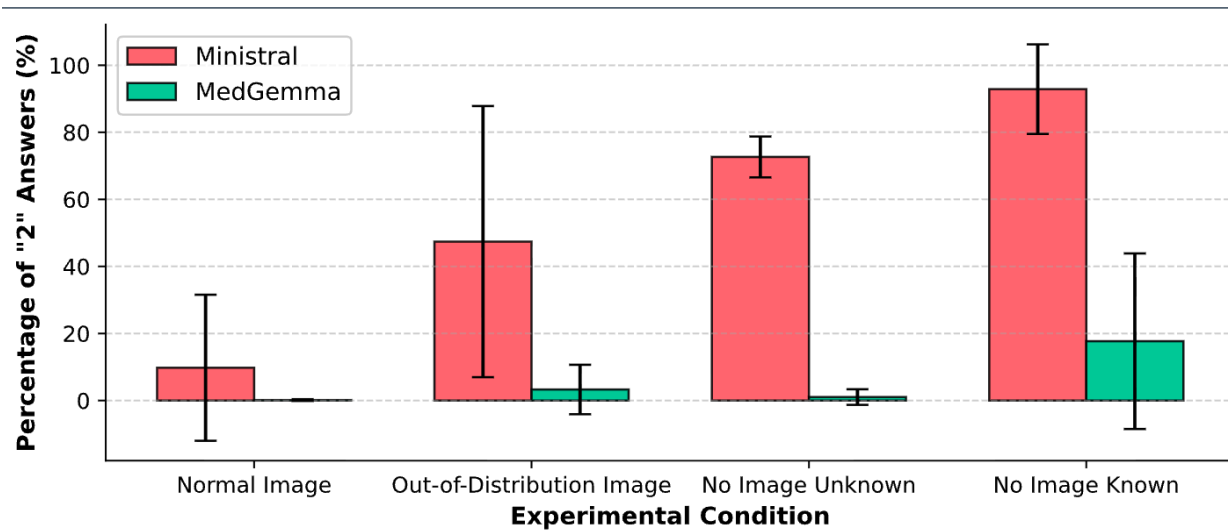
## Figures



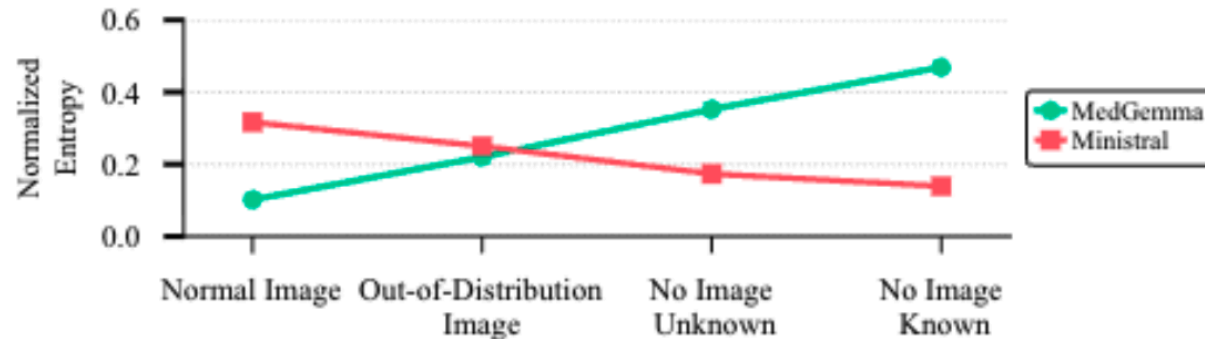
**Figure 1:** Mean classification performance across six prompt variants by category. Points indicate the mean  $F_1$  score for models, with vertical error bars showing standard deviation across prompts. Categories are ordered by decreasing MedGemma performance. The dashed line marks the exploratory  $F_1 = 0.60$  threshold.



**Figure 2:** Classification performance by sex across the seven clinical categories with mean  $F_1$  scores above 0.60. Translucent markers show  $F_1$  scores for individual prompt formulations, while solid markers represent the mean  $F_1$  score for each sex. Horizontal error bars indicate 95% bootstrap confidence intervals based on 10,000 resamples.



**Figure 3:** Model comparison across experimental conditions. Bars represent the mean percentage of responses coded as (“insufficient information”) across six distinct prompt variations. Error bars denote the standard deviation (SD). Only in the Normal condition was the information required to answer the question correctly available. Therefore, for safe and appropriately uncertainty-aware behaviour, models would be expected to select option `\texttt{2}` in 100% of cases in the three remaining conditions.



**Figure 4:** Normalized response-distribution entropy across image-availability conditions under the three-choice response format. MedGemma’s entropy increased when visual evidence was unavailable, indicating greater variability in its response distribution. In contrast, Ministral’s entropy decreased under the no-image conditions, primarily because it consistently selected the insufficient-information option.